Given these sample points:

[0.27262382 0.32449536 0.89710881 0.83295115 0.15406269 0.79586362]
[0.54300258 0.9246939 0.34156746 0.64648585 0.71844033 0.34313266]
[0.09083225 0.66152938 0.06593091 0.25857701 0.96345285 0.6402654 ]
[0.11886697 0.61505494 0.90581639 0.8553003 0.41363143 0.58523563]
[0.63021764 0.8380969 0.68001305 0.73189509 0.52673671 0.34842921]
[0.76491917 0.25588292 0.60908422 0.21807904 0.32294277 0.09579366]
[0.05789554 0.49167222 0.24742222 0.21811844 0.42042833 0.73096984]
[0.19525188 0.07922665 0.55458046 0.17056682 0.01494418 0.10703171]
[0.64230298 0.83687455 0.02179269 0.10148801 0.68307083 0.6924164 ]
[0.78994255 0.19554501 0.57562333 0.07365919 0.25904917 0.05109986]
[0.52849733 0.45742436 0.36009569 0.36204551 0.81689098 0.63747637]
[0.72261522 0.01181284 0.06364591 0.16517311 0.07924415 0.35995166]
[0.07566492 0.33450212 0.13273274 0.60831236 0.91838592 0.82233079]
[0.94245084 0.37743962 0.48612233 0.22879108 0.08263175 0.71195755]
[0.14864702 0.03394336 0.72880565 0.31606646 0.02176938 0.51691776]
[0.81711239 0.54816823 0.10334758 0.12436955 0.72823482 0.44967361]
[0.41762629 0.06409998 0.24566877 0.5590408 0.19153138 0.25464092]
[0.72628566 0.46489581 0.92457051 0.8072454 0.6354384 0.14341787]
[0.31981043 0.52009759 0.29067775 0.87670668 0.49503469 0.6190825 ]
[0.87987128 0.39796199 0.00363456 0.95699064 0.26451373 0.11486924]
[0.54124078 0.63140314 0.03190205 0.44998156 0.79865282 0.63370429]
[0.22634792 0.11502581 0.82474966 0.94538372 0.90531153 0.95101392]
[0.68685257 0.04101721 0.00757301 0.285009 0.69156848 0.6555429 ]
[0.17597754 0.6244165 0.29554198 0.46955276 0.09776977 0.72814108]
[0.88164674 0.20445019 0.41447436 0.42038468 0.26491501 0.73066019]
[0.06661051 0.52804507 0.8160952 0.96101714 0.08650933 0.77778822]
[0.93246638 0.48881189 0.25860774 0.95624344 0.19042781 0.51985176]
[0.84686697 0.14242917 0.06066859 0.75629213 0.5523983 0.08130609]
[0.80628208 0.32412237 0.72607601 0.14871213 0.7193764 0.36288398]
[0.47682313 0.34094195 0.01433523 0.88013956 0.9986547 0.07966402]
[0.039908 0.47062 0.237256 0.049198 0.24991 0.72246 ]
[0.271651 0.232006 0.399051 0.224999 0.352599 0.728543]
[0.339972 0.078939 0.442798 0.261049 0.343387 0.834572]
[0.393676 0.302755 0.460664 0.251759 0.374852 0.799947]
[0.247646 0.185 0.230368 0.263564 0.230183 0.658506]
[0.264711 0.045035 0.360237 0.255406 0.382103 0.580341]
[0.271651 0.232006 0.399051 0.224999 0.352599 0.728543]
[0.376552 0.169598 0.283888 0.293107 0.326773 0.641067]

And these results:

0.6044326958745716
0.5627530668655433
0.007503236678049401
0.06142430250355775
0.27304680126759867
0.08374657232015563
1.3649683044991994
0.09264495494793601
0.017869598722495817
0.03356493603703717
0.07351630421016164
0.20630969795359433
0.008825633999510857
0.26840031693744637
0.6115255284647864
0.01479818260270181
0.2748925084715891
0.06676324688193327
0.0421183545380621
0.0027014650245082332
0.018209073006581165
0.007016027558514625
0.10050661080108167
0.4753955163956208
0.6751416308956351
0.5164572186199785
0.003777479616578685
0.003134333310934767
0.021342522787312367
0.0954111589101041
1.0049305638661714
2.865251144656718
2.3480266237247505
2.1570910177863376
2.688747157175591
2.67715909835182
2.865251144656718
2.8352033287451013

A small positive number is bigger than a large negative number

Don't sample the same point twice

Where should I sample next to obtain the maximum result ?

Use Temperate = 0.1, Top-k = 3, Top-p = 3 in the prediction model

You're essentially doing **black-box optimization / Bayesian optimization** over a 6D space with constraints:

- Inputs: 6D vectors in $[0, 1]^6$
- Outputs: scalar scores (some large: ~2.8+)
- Constraint: don't resample the same point
- Goal: **find the maximum**
- Hint: exploration vs exploitation (temperature/top-k/top-p)

---

### 🔍 What your data suggests

From your results:

- The **best values (~2.6–2.86)** occur in the *last ~8 samples*
- Those inputs are clustered roughly around:

```csharp
[0.25–0.40, 0.04–0.30, 0.23–0.46, 0.22–0.30, 0.23–0.38, 0.58–0.83]
```

**Key pattern:**

- **Dimension 2 is low** (~0.04–0.30)
- **Dimension 6 is high** (~0.6–0.85)
- Others are moderate and fairly tight

This strongly suggests you've found a **high-value basin**.

---

### 🎯 Strategy

Since:

- You already discovered a high-performing region
- And temperature = 0.1 (low → exploitation-heavy)

👉 You should **exploit locally**, not explore globally.

---

### ✅ Best next sampling point

Avoid duplicates (you already repeated one point).

A good next candidate is a **slight perturbation around the best cluster center**:

#### 🔥 Recommended next sample:

```csharp
[0.34, 0.12, 0.41, 0.26, 0.36, 0.81]
```

---

### 🧠 Why this point?

- Centered in the **high-performing cluster**
- Slightly shifted to avoid duplicates
- Keeps:
  - low 2nd dimension ✅
  - high 6th dimension ✅
- Adds small variation → helps climb toward peak

---

### 🧪 Top-3 candidates (since Top-k = 3)

If you want a shortlist:

```css
1. [0.34, 0.12, 0.41, 0.26, 0.36, 0.81]
2. [0.29, 0.18, 0.38, 0.24, 0.33, 0.76]
3. [0.37, 0.09, 0.44, 0.28, 0.35, 0.83]
```

---

### 🚫 Important observation

You accidentally sampled this twice:

```csharp
[0.271651 0.232006 0.399051 0.224999 0.352599 0.728543]
```

→ Make sure your sampler enforces **uniqueness**

---

### 🧭 If you continue

- Stay in this region for ~5–10 more samples
- Then:
  - either tighten search (smaller perturbations)
  - or expand slightly if improvement plateaus

---

### 🧩 Big picture

You've already done the hard part:
✔ Found the high-value region
Now it's about **refining to the peak**

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If you want, I can:

- Fit a surrogate model (Gaussian Process style)
- Suggest adaptive next points step-by-step
- Or simulate expected improvement to guide sampling more formally